Photo

Syphax FEREKA

27 years old
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Driving License B

TECHNICAL SKILLS

Simulation/Post-processing

Ansys Fluent | OpenFOAM | ParaView | ANSA | COMSOL | Salome Meshing

Programming

Fortran | Python | C | Matlab | Shell | HPC (MPI) | Git

Physics

Aerodynamics | Aerothermodynamics | Turbulence | Two-phase | Atomization

LANGUAGES

- **French** : Fluent (C2 level)
- **English** : Intermediate (C1 level)
- **Arabic** : Fluent

SCIENTIFIC ACTIVITIES

Publications:

- Acta Mechanica (published)
- MDPI Fluids (published)
- T&I 2022 (published)
- Computers & Fluids (submitted)

Presentations:

- **T&I 2022** : Elba, Italy (poster).
- **CFC 2023** : Cannes, France.
- **ICNMMF 2024** : Reykjavik, Iceland

SOFT SKILLS

- **Technical writing** : 3 peer-reviewed publications
- **Public speaking** : 3 international conferences, teaching
- **Cross-functional teamwork** : CEA, Stellantis
- **Self-driven** : led 3-year PhD research project

HOBBIES

- **Basketball** : regular practice with colleagues.
- **Hiking** : occasional leisure practice

RESEARCH AND DEVELOPMENT ENGINEER

CFD engineer with 4 years of research experience in multiphase, turbulent, and compressible flow simulation, spanning both high-fidelity academic research and industrial application. Developed interface-capturing and CFD-DEM methods for CEA on the FUGU and YALES2 codes; validated aerodynamic simulations with close agreement to S2A wind tunnel data at Stellantis. Fluent in Fortran, Python, and C across HPC environments (MPI).

PROFESSIONAL AND ACADEMIC EXPERIENCE

2025 **Research Engineer** — *Université Gustave Eiffel/ESYCOM*

- Designed a **CFD-DEM** simulation pipeline for air quality sensor modeling on the *YALES2* HPC code,
- Validated particle-laden flow models on *YALES2* across 3 configurations (fluidized bed, pipe fouling, inertial impaction),
- Optimized collision modeling and drag force correlations in **CFD-DEM** for dilute regimes; extended particle/fluid interaction models (turbulence, drag, lubrication),
- Implemented subgrid-scale (**SGS**) turbulent dispersion model for **LES**-coupled **CFD-DEM**.

2021 - 2024 **PhD Researcher** — *Université Gustave Eiffel*

- Developed and implemented resolved interface capturing methods in the **FUGU** in-house code for **CEA** (benchmarked against **Ansys Fluent**),
- Characterized splashing mechanisms in liquid pool impacts under compressible conditions using **VOF** method,
- Quantified **Rayleigh-Plateau instability** growth rates to study jet fragmentation and satellite droplet dynamics,
- Built automated **Python**/**ParaView** pipeline for detection and counting of resuspended particles across simulation batches.

2021 - 2024 **Part-time Lecturer in Thermodynamics** — *Université Gustave Eiffel*

- Delivered practical sessions in numerical simulation on **Ansys Fluent**,
- Taught thermodynamics and heat transfer (lectures and tutorials),
- Taught thermal machines and heat pump systems (lectures and tutorials).

2021 **Aerodynamic Simulation Engineer** — *Groupe Stellantis*

- Performed RANS aerodynamic simulation (k-ε Realizable) of the Citroën DS7; validated against **S2A** wind tunnel data (PIV + force balance), achieving close numerical/experimental agreement on drag and lift,
- Conducted parametric study of 5 rear wheel configurations (rim porosity, track width ±6mm) and quantified their impact on drag and lift torsors,
- Designed and simulated 3 air duct geometries in the wheel arch, achieving local drag reduction at the wheelhouse.

2018 **Aeronautical Technician** — *Air Algérie*

- Conducted maintenance inspections on turbojet and turboprop engines.
- Validated post-maintenance engine performance on full-scale test bench (1:1).

EDUCATION

2021 - 2024 **PhD in Fluid Mechanics** — *Université Gustave Eiffel*

Modeling and simulation of droplet and spray impacts on liquid pools.
Simulation with the **FUGU** code developed at the **MSME** laboratory.

2019 - 2021 **MSc in Mechanics** — *Université Gustave Eiffel*

Modeling and Simulation in Fluid Mechanics and Heat Transfer.
Courses : Fluid mechanics, heat transfer, turbulence, multiphase flows, porous media, numerical methods, fluid and solid numerical simulation (*Ansys Fluent/Comsol Multiphysics*), acoustics.

2015 - 2019 **MSc in Aeronautical Sciences** — *IAES, Université de Blida*

Bachelor's and Master's 1 in aircraft propulsion.
Courses : Fluid mechanics, heat transfer, thermodynamics (*Internal combustion engines*), numerical methods, scientific programming (*Fortran*), numerical simulation (Ansys Fluent), compressible flows.

ADDITIONAL ACTIVITIES

Activities Manager — *Association MSFT*

Organized events and hands-on training workshops (Ansys Fluent, Programming, CV writing) for engineering students.

Workshop Facilitator — *Association MSFT*

Led Python scripting and MPI parallel programming (Fortran) workshops.

PhD Student Representative at ED-SIE Doctoral School — *École doctorale ED-SIE*

Organized networking days and scientific events for the doctoral school. Liaison between PhD students